



Random Forest • Live Dataset Analysis

Fraud Detection Analytics Dashboard

Monitor transaction activity, identify fraud patterns, analyze hourly behavior, and evaluate the performance of the machine learning fraud detection model.

Machine Learning • Random Forest Classifier



Transaction Summary

Overall transaction and fraud statistics

TOTAL TRANSACTIONS



284807

Transactions analyzed

FRAUDULENT TRANSACTIONS



492

Model predicted fraud records

NORMAL TRANSACTIONS



284315

Model predicted normal transactions

FRAUD RATE



0.17%

Percentage of transactions



Transaction Amount Analysis

Financial distribution across the dataset

TOTAL TRANSACTION AMOUNT

₹25162590.01

Combined value of all transactions

FRAUD TRANSACTION AMOUNT

₹60127.97

Combined value of predicted fraudulent transactions

AVERAGE TRANSACTION AMOUNT

₹88.35

Average amount per transaction



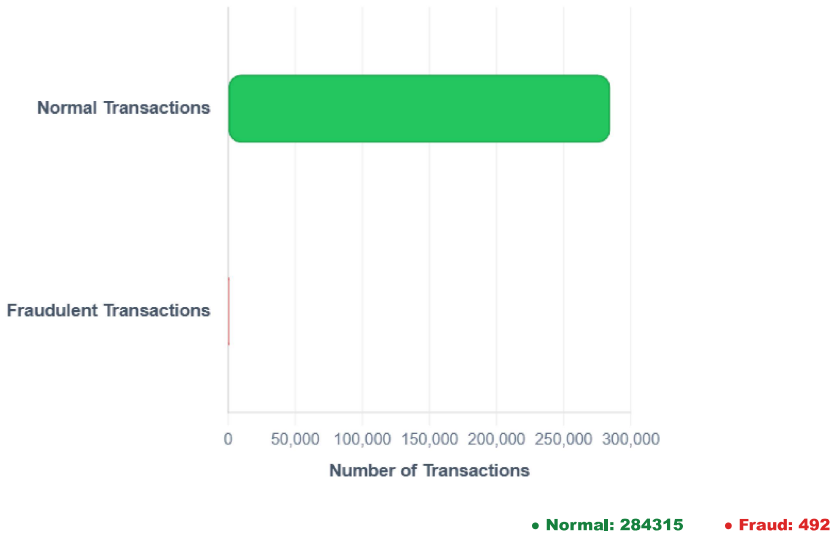
Fraud Analytics

Visual analysis of fraud distribution and activity



Fraud vs Normal Transactions

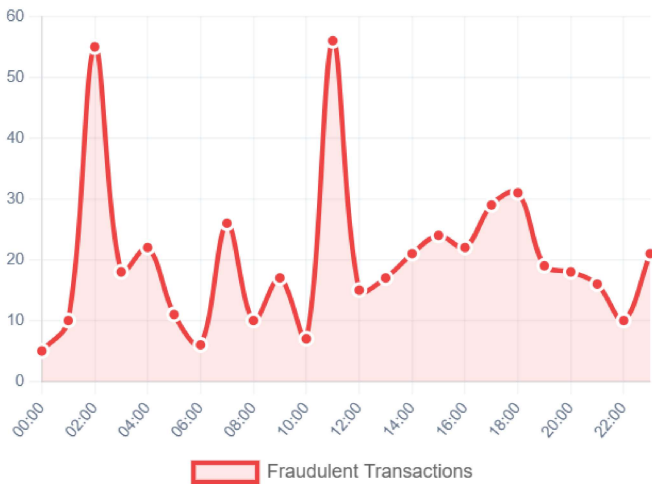
Model-predicted normal and fraudulent transaction counts



Fraud Activity by Hour

Number of fraudulent transactions across 24 hourly periods

🔥 Peak Fraud Hour: 11:00 (56 transactions)



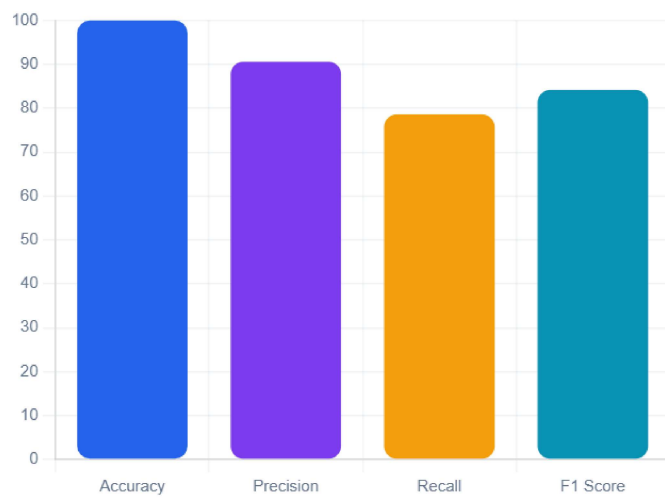
Average Amount Comparison

Average transaction amount for normal and fraudulent transactions



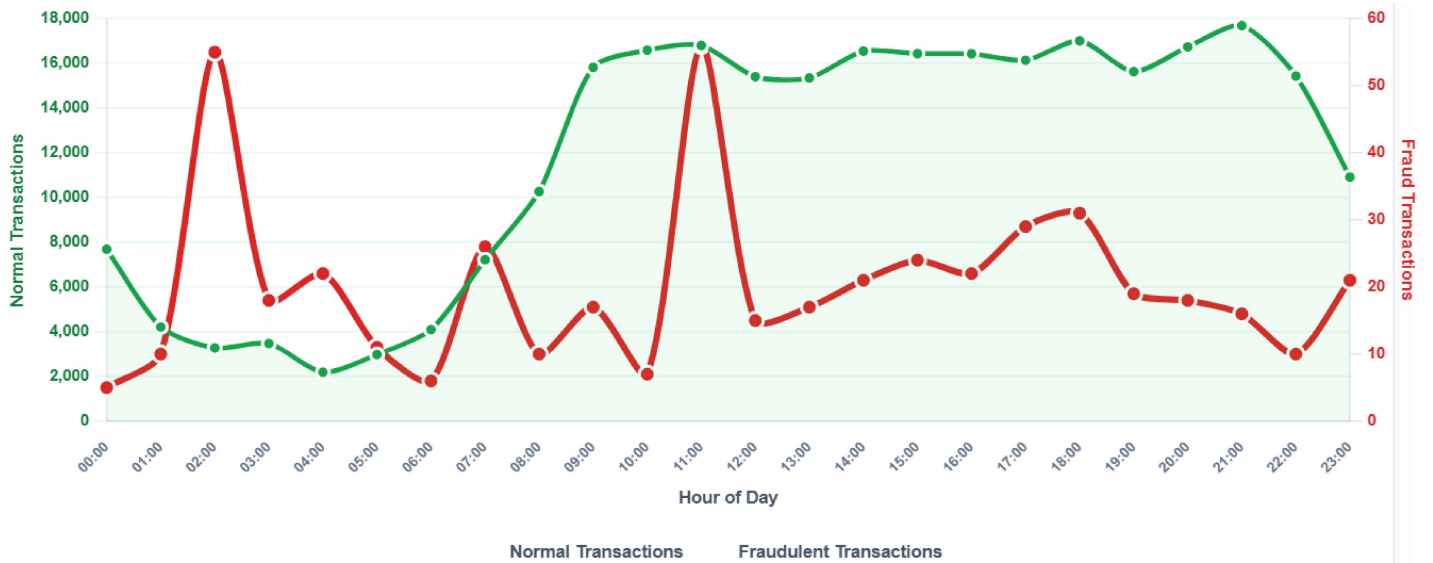
Model Performance

Performance metrics of the Random Forest classifier



Normal vs Fraud Transactions — 24-Hour Comparison

Compare normal and fraudulent transaction activity for each hourly bucket



Detection Performance Metrics

Key evaluation metrics for the fraud detection model

ACCURACY

99.95%

Overall correct predictions

PRECISION

90.59%

Correct fraud predictions

RECALL

78.57%

Fraud cases identified

F1 SCORE

84.15%

Precision-recall balance



Classification Results

Confusion matrix showing how the Random Forest model classified normal and fraudulent transactions

Actual \ Predicted

Normal

Fraud

Normal

284300

TRUE NEGATIVE
Normal → Normal

15

FALSE POSITIVE
Normal → Fraud

Fraud

21

FALSE NEGATIVE
Fraud → Normal

471

TRUE POSITIVE
Fraud → Fraud

TN

Normal correctly identified

FP

Normal incorrectly classified as fraud

FN

Fraud incorrectly classified as normal

TP

Fraud correctly identified



About Fraud Detection Model

Machine learning approach used by Fraud Intelligent



Random Forest Classifier

Fraud Intelligent uses a Random Forest machine learning classifier to analyze transaction characteristics and identify potentially fraudulent activity. The system evaluates transaction features, generates predictions, calculates fraud probability, and assigns a risk level to help users understand suspicious transactions.

Random Forest

Fraud Detection

Probability Analysis

Risk Assessment

24-Hour Analytics

Batch Prediction



Fraud Intelligent System • Machine Learning Fraud Detection